
What is a Macro in Excel?

A **Macro in Excel** is a set of instructions written in **VBA (Visual Basic for Applications)** that automates repetitive tasks.

Instead of performing the same steps again and again manually, you can record or write a macro to do it for you with a single click or shortcut key.

For example:

- If you always format a report (change fonts, apply colors, adjust column width), you can record those steps once as a macro and run it whenever needed.

Why do we use Macros in Excel?

Macros are mainly used for **automation** and **efficiency**.

Save Time – Automates repetitive tasks like formatting, calculations, or report generation.

Reduce Errors – Eliminates mistakes caused by manual work.

Consistency – Ensures the same steps are followed every time.

Complex Calculations – Perform calculations or tasks that are difficult to do manually.

Custom Solutions – Create your own mini-programs for business needs.

Example of Macro Usage

1. Recording a Macro:

- Suppose every day you receive a sales report.
- You want to **bold headers, adjust column width, and apply filters**.
- Instead of repeating these steps, you can record a macro once.

2. Running the Macro:

- Next time, just click the macro button (or press a shortcut like Ctrl + Shift + S) and Excel will do all steps automatically.

Syntax (Basic VBA Macro Example)

Here's a simple VBA macro code:

```
Sub FormatReport()
```

'This macro formats the report

Range("A1:D1").Font.Bold = True

Range("A1:D1").Interior.Color = RGB(200, 200, 250)

Columns("A:D").AutoFit

End Sub

This macro will bold the first row, apply a background color, and autofit columns A to D.

In short:

A **Macro** = A tool in Excel to **automate repetitive tasks** using **VBA programming**.